

Discretion versus Policy Rules in Practice

John B. Taylor

PUBLICATION YEAR	1993
JOURNAL	Carnegie-Rochester Conference Series on Public Policy, Vol. 39, pp. 195–214 (North-Holland / Elsevier)
DOI	10.1016/0167-2231(93)90009-L
RESEARCH AREA	Monetary Economics – Macroeconomic Policy Rules
SERIES	Monetary Economics
PAPER TYPE	Mixed – applied policy paper combining an illustrative calibrated model with narrative case-study evidence
REVIEWED BY	Arthur Faugeron
DATE REVIEWED	29 August 2026

ABSTRACT

This paper addresses a specific design problem left open by the rules literature of the 1980s. Research on rational-expectations policy evaluation had converged on the desirability of systematic interest-rate rules that respond to inflation and output, but had said little about how a central bank could operate such a rule once mechanical, period-by-period commitment to a fixed algebraic formula proved practically impossible. Taylor's response is to separate the design, transition, and operation of a policy rule, and to propose a single illustrative reaction function, later known as the Taylor rule, in which the federal funds rate responds to output and inflation gaps around a two percent real-rate benchmark. The rule is not econometrically estimated: it is calibrated with round coefficients and then checked, informally, against the actual path of the federal funds rate between 1987 and 1992, and stress-tested through narrative discussion of two 1990 episodes, the Iraqi oil-price shock and German unification, in which incoming information appeared to call for deviation from the rule's literal prescription. The paper's central claim is that rule-like behavior, understood as a systematic and credible commitment to a reaction pattern rather than a mechanically applied formula, describes recent Federal Reserve conduct well and should guide, without dictating, future policy decisions.

1. Introduction

By the early 1990s, the academic literature on monetary policy had accumulated a set of results that pointed, almost unanimously, toward one conclusion: policy rules that respond systematically to observable macroeconomic conditions outperform discretionary, period-by-period decision-making. This conclusion rested on several distinct strands of research that had matured over the preceding fifteen years. The time-inconsistency literature, associated with PAPER Kydland and Prescott (1977) and formalized in reputational terms by Barro and Gordon (1983), had shown that a discretionary policymaker with an incentive to exploit a short-run inflation-output trade-off generates a socially costly inflation bias that a credible, pre-committed rule avoids. The Lucas critique had shown that traditional reduced-form policy evaluation is invalid once private agents' expectations are allowed to respond to the policy regime itself, which pushed empirical work toward structural, rational-expectations models in which policy-rule parameters could be changed and the model re-solved for a new equilibrium. And a growing body of multicountry, rational-expectations simulation work, most systematically compiled in the forthcoming Bryant, Hooper and Mann (1993) volume, had begun to produce a rough professional consensus about the functional form of good monetary policy rules: short-term interest rates should respond to deviations of the price level (or inflation) and real output from target, rather than to money-supply or exchange-rate targets.

What this literature had not done, in Taylor's own diagnosis, is provide a bridge from the academic conclusion to the practice of a committee such as the Federal Open Market Committee (FOMC), which meets periodically, faces genuinely novel shocks, and cannot credibly commit to executing a fixed algebraic formula every quarter. The paper is best classified as a mixed, applied policy paper: it is not a formal econometric exercise in its own right, and it does not present new theory. It is a conceptual and illustrative piece that operationalizes results Taylor had developed with an estimated rational-expectations multicountry model in a separate, non-reproduced body of work (Taylor, 1979; Taylor, 1993, the book *Macroeconomic Policy in a World Economy*), combined with a narrative examination of how a hypothetical rule would have been used, or deviated from, during two concrete episodes in 1990. The paper's lasting significance is almost entirely due to a single equation, presented without derivation or estimation, that became known as the Taylor rule and subsequently entered the standard toolkit of academic monetary economics and, decades later, the Federal Reserve's own public communication.

2. Research Problem and Hypothesis

2.1 Research Question

Given that rational-expectations, multicountry model simulations had identified interest-rate feedback rules of the general form "raise the short-term rate when inflation or output rise above target" as delivering lower joint output-price variability than money-supply or exchange-rate-

based rules, can the systematic, rule-like content of such a policy be preserved and made operationally useful for a central bank that cannot commit to mechanically executing a single algebraic formula period by period, and does a simple illustrative version of such a rule in fact resemble the Federal Reserve's own recent conduct?

2.2 Motivation

The rules-versus-discretion question is one of the oldest positive and normative questions in monetary economics, and by 1993 it had reached an unusual state: theory had a clear answer (rules dominate discretion when policymakers cannot resist short-run temptation), but policy practice showed almost no visible trace of that answer. Central bankers did not, and showed no sign of beginning to, follow any published algebraic rule. The paper matters because it is the specific point at which the abstract credibility literature entered central-bank practice in a form concrete enough to be used: the illustrative equation proposed here is the direct ancestor of the "Taylor rule" that is used today, by the Federal Reserve itself, the European Central Bank, the International Monetary Fund, and financial-market economists, as a standard benchmark for judging whether a policy rate is too high or too low relative to macroeconomic conditions.

2.3 Existing Literature

Three bodies of prior work frame the paper. First, the time-inconsistency and credibility literature (Kydland and Prescott, 1977; Barro and Gordon, 1983; and, in textbook form, Blanchard and Fischer, 1989) established formally why a policy rule, understood in that literature as the "optimal," "rules," or "precommitted" solution to a dynamic optimization problem, dominates the "inconsistent," "cheating," or "shortsighted" discretionary solution, in the same sense that a cooperative outcome dominates a noncooperative one in game theory. Second, Taylor's own prior structural work (Taylor, 1975, on transitions to rational expectations; Taylor, 1979, on estimation and control of a rational-expectations macroeconomic model; and the companion 1993 book) supplies the estimated multicountry model whose simulated rankings of alternative policy rules this paper summarizes. Third, McCallum (1988) had proposed a rival simple rule based on the growth rate of the monetary base, representing an alternative strand within the same broader research program that this paper acknowledges but does not test against its own proposed rule.

2.4 Research Gap

None of this literature had translated "a rule performs well in model simulation" into a workable description of how a real committee should use a rule when quarterly data are noisy, structural breaks and one-off shocks are common, and mechanical commitment to a formula is neither credible nor, in the paper's own account, desirable. That operational gap, rather than any unresolved question about the design of the rule itself, is what the paper targets.

2.5 Contribution

The paper's contribution has three parts: a semantic clarification distinguishing the design of a policy rule, the transition to a new rule, and the day-to-day operation of an existing rule; a single, deliberately simple illustrative rule, equation (1) below, calibrated with round coefficients and shown, by visual comparison, to track recent Federal Reserve behavior; and two narrative case studies illustrating how a policymaker who treats the rule as one input among several, rather than as a mechanical instruction, would reason through genuinely novel shocks. **CRITIQUE** Whether the second contribution is actually demonstrated, in the sense of being tested, rather than merely illustrated by a single non-statistical plot, is doubtful; this is developed in Sections 6 and 7 below.

3. Theoretical Framework and Mechanism

The paper's single formal object is an illustrative interest-rate reaction function, presented without derivation from an optimization problem or an estimated equation:

$$r = p + 0.5y + 0.5(p - 2) + 2 \quad (1)$$

r the federal funds rate (percent, target level).

p the rate of inflation over the previous four quarters (percent).

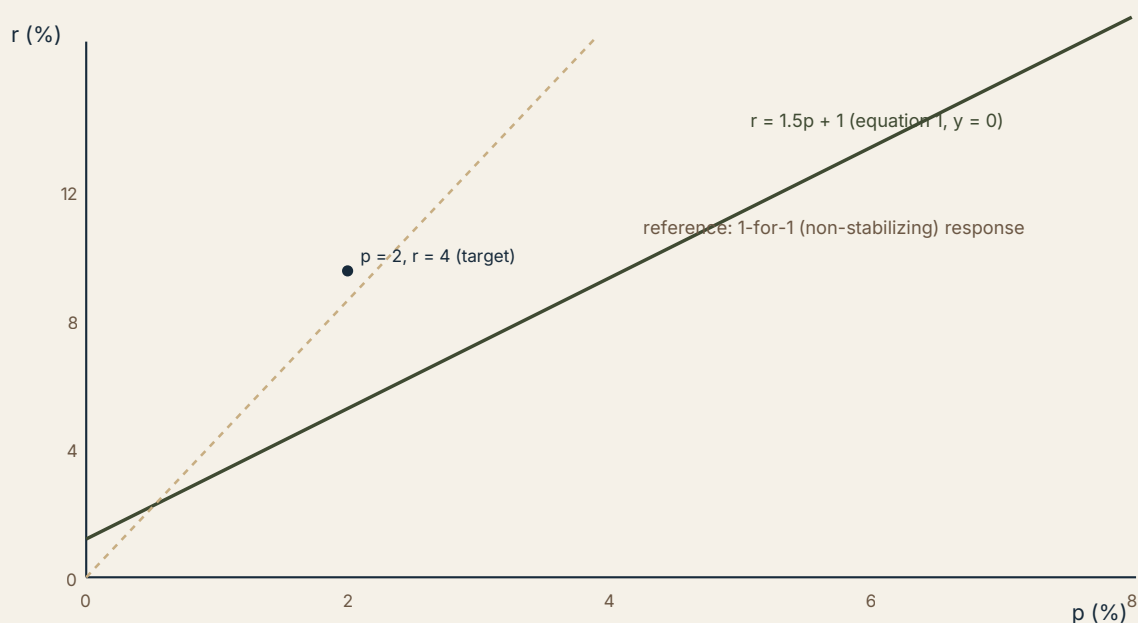
y the percentage deviation of real GDP from trend, $y = 100(Y - Y^*)/Y^*$, where Y is real GDP and Y^* is trend real GDP, assumed by the author to grow at 2.2 percent per year over 1984:1–1992:3.

Collecting terms gives the equivalent, and more transparent, form $r = 1.5p + 0.5y + 1$. Three features of this rewritten form carry the paper's economic content. First, the coefficient on the inflation gap is 1.5 rather than 1: a one-percentage-point rise in inflation raises the nominal funds rate by one and a half points, so the *real* ex post short-term rate ($r - p$) rises by half a point whenever inflation rises. **INFERENCE** This property, that the nominal rate must move more than one-for-one with inflation for the rule to be stabilizing, is exactly what later literature (Clarida, Gali and Gertler, 2000; Woodford, 2001) would formalize as the condition for a determinate, non-explosive rational-expectations equilibrium in a New Keynesian model, a condition now generally called the Taylor principle. The 1993 paper itself does not derive or even state this condition; it is discovered, ex post, in the algebra of a rule Taylor proposed on grounds of round numbers and rough consistency with simulation and historical fit, not on grounds of equilibrium determinacy.

Second, the constant term of 1 (equivalently, the "2" that appears directly in equation 1) sets the rate that would prevail if both gaps were closed: with $p=2$ and $y=0$, equation (1) gives $r=4$ percent nominal, or 2 percent in real terms. The paper states that this 2 percent equilibrium real rate is "close to" the assumed 2.2 percent steady-state growth rate, but this correspondence is asserted rather than derived from an intertemporal Euler condition; unlike the Fisherian steady-state relationship that would later be built explicitly into New Keynesian models, here it is simply a plausibility check on a chosen constant.

Third, both response coefficients are set to 0.5, described explicitly by the author as "round numbers that make for easy discussion," not as the output of any estimation procedure applied within this paper. The rule is a *calibrated*, illustrative object, not an *estimated* reaction function; the distinction matters because it means the equation cannot, by construction, come with any standard error, confidence interval, or goodness-of-fit statistic. The paper further specifies that p is measured as trailing four-quarter inflation, explicitly described as "a proxy for expected inflation," an assumption of adaptive rather than genuinely forward-looking expectations that sits uneasily beside the rational-expectations framework invoked to justify the underlying multicountry simulation evidence discussed in Section 4.

Figure 1. Equation (1) evaluated at trend output ($y = 0$): the federal funds rate implied by the rule as a function of the inflation gap



Reviewer's own construction from equation (1) as stated in the paper, holding the output gap at $y = 0$. Not one of the paper's original figures; the paper's own Figures 1–3 plot the 1987–1992 time series discussed in Section 5 and are not digitized here because the source copy consulted did not yield machine-readable axis values. Source: author's calculation.

The figure makes visible what the algebra already shows: because the solid line has a slope of 1.5, greater than the dashed 45-degree reference line of slope 1, every increase in inflation is met with a more-than-proportional increase in the nominal rate, so the real rate implied by the rule is increasing in the inflation gap. This is the qualitative content later formalized as the Taylor principle, present in the paper's arithmetic but never named, tested for, or connected to equilibrium determinacy within the paper itself.

Finally, it is important to be explicit about what this paper does *not* contain. The structural, estimated rational-expectations multicountry model that generates the quantitative simulation claims discussed in Section 4 is not reproduced here in any equation form; it is cited to Taylor (1993, the book) and is, from the standpoint of this conference paper alone, a black box. **CRITIQUE A** reader of this paper cannot verify, reconstruct, or stress-test the model that supposedly grounds the claim that price/output interest-rate rules dominate money- or exchange-rate rules; that evidentiary burden is placed entirely on a separate, unreproduced monograph.

4. Empirical and Analytical Basis

The paper does not present a dataset, a regression, or a data appendix in the conventional empirical sense. Its evidentiary base has two distinct layers, and it is important to keep them separate because they carry very different evidentiary weight.

4.1 Cross-model simulation evidence (summarized, not reproduced)

The first layer is qualitative summary of results from two bodies of simulation work neither of which is reproduced numerically in this paper: the nine-model comparison compiled in Bryant, Hooper and Mann (1993), built by the International Monetary Fund, the Federal Reserve Board, Canada's Department of Finance, and individual researchers, seven of the nine models being estimated rational-expectations systems; and Taylor's own multicountry rational-expectations model (Taylor, 1993, the book), simulated under alternative monetary policy rules for the G-7 economies. The "identification" strategy behind both exercises is standard to the rational-expectations policy-evaluation literature that responded to the Lucas critique: rather than regressing observed policy instruments on outcomes, which the Lucas critique shows is invalid once expectations respond to the policy regime, the exercise changes the policy-rule parameters embedded in an estimated structural model and resolves the model for a new rational-expectations equilibrium, then compares the resulting output and price variability across rule regimes under an empirically estimated distribution of historical shocks. This is a legitimate and by-1993 well-established identification strategy for counterfactual policy comparison, but it is only as credible as the structural model in which it is embedded, and that model is, again, external to this paper.

4.2 U.S. time series, 1987–1992

The second layer is an actual, if small, empirical exercise: the federal funds rate, the GDP-deflator inflation rate over the trailing four quarters, and real GDP relative to an assumed 2.2 percent annual trend from 1984:1, plotted quarterly from 1987 to 1992 and compared informally, by eye, to the path equation (1) would have prescribed. No source citation, vintage, or construction method is given for the 2.2 percent trend-growth assumption; it is simply asserted. **CRITIQUE** This is a genuine data-transparency gap: a reader cannot tell whether the trend was estimated by a linear filter, a structural potential-output model, or eyeballing, and the entire output-gap term of equation (1) depends on it.

4.3 Identification in the two case studies

For the two 1990 case studies discussed in Section 5, the paper's "identifying variation" is narrative rather than statistical: the slope of the oil futures curve is used to infer that market participants expected the 1990 oil-price shock to be temporary, and the co-movement of German and U.S. long-term bond yields is used to infer that the early-1990 rise in U.S. long rates reflected a common, real, cross-border shock (anticipated German unification-driven investment demand) rather than a U.S.-specific rise in inflation expectations. Both inferences are plausible and are supported by

directionally correct auxiliary evidence quoted in the text, but neither is subjected to a formal test; both are, in the paper's own presentation, expert narrative judgment rather than an estimated model.

5. Results

5.1 Cross-model and multicountry simulation findings

The paper reports, without a reproduced table, that policy rules focused on the exchange rate or on the money supply deliver worse output and price stability than policy rules that focus directly on the price level and real output, and that this finding holds with "some consensus" across the nine independently built models compared in Bryant, Hooper and Mann (1993), despite substantial disagreement across those models about the exact coefficients. From Taylor's own multicountry model, comparing a flexible-exchange-rate regime (each country targets its own price level and output with its own interest rate) against a fixed-exchange-rate regime (a common "world" interest-rate target), the paper states specific, if qualitative, magnitudes.

Table 1. Qualitative comparison of macroeconomic volatility, fixed versus flexible exchange-rate regimes, as reported in the text

Country / bloc	Output volatility, fixed vs. flexible	Price-level volatility, fixed vs. flexible
Germany	Standard deviation of output nearly doubles under fixed rates	More than double under fixed rates
Japan	Standard deviation of output nearly doubles under fixed rates	More than double under fixed rates
United States, France, Italy, United Kingdom	Larger under fixed rates (magnitude not separately quantified in the text)	Greater under fixed rates in all countries
Canada	Slightly lower under fixed rates than flexible	Deteriorates under fixed rates despite the output result

Source: qualitative statements in Taylor (1993), main text; the paper does not report the underlying numerical standard deviations, only these comparative descriptions.

The paper draws two further qualitative conclusions from this exercise: it is preferable for each central bank to set its interest rate based on domestic conditions rather than the exchange rate, and placing positive weight on both the price level and real output in the interest-rate rule outperforms a rule that responds to the price level alone in most countries, although the paper explicitly states that it is not clear whether the weight on output should exceed or fall short of the weight on inflation.

5.2 The illustrative rule against 1987–1992 U.S. policy

The paper's Figures 1 through 3 (not digitized in this review; see the note under Figure 1 above) plot, respectively, the actual and rule-implied federal funds rate, real GDP against its 2.2 percent trend, and trailing four-quarter GDP-deflator inflation, each quarterly from 1987 to 1992. The paper's own reading of these figures is that the rule-implied path tracks the actual funds rate "remarkably well," with one significant divergence in 1987, when the Federal Reserve eased policy in response to the October 1987 stock-market crash, an event with no counterpart in equation (1)'s two arguments. The output-gap figure shows the economy running above trend in the late 1980s and falling below trend during the 1990–91 recession, with only a slight narrowing of the gap by the end of the sample; the inflation figure shows the disinflation that equation (1) is implicitly responding to over the same period. PAPER Taylor's own conclusion is that "it certainly appears that the changes in inflation and real GDP influenced the path of the federal funds rate," a carefully hedged causal claim that stops short of asserting the Federal Reserve was in fact using anything like equation (1).

5.3 Case study one: the 1990 oil-price shock

Iraq's invasion of Kuwait on 2 August 1990 removed roughly 4.3 million barrels per day of combined Iraqi and Kuwaiti oil production from the market and threatened Saudi supply. The spot oil price rose from 21 dollars per barrel at the end of July to 28 dollars by 6 August and peaked near 46 dollars in mid-October; the monthly average rose from 17 dollars in July to 36 dollars in October. A Council of Economic Advisers consensus estimate, quoted in the paper, held that a one-year temporary 50 percent oil-price increase would raise the GDP deflator by about 1 percent temporarily and, with a longer lag, reduce real output by a comparable amount. Applied literally, equation (1) would call for a funds-rate increase in the short run, since the price-level effect arrives before the output effect. The paper argues this literal prescription would have been the wrong response, and supports the argument with the one-year-ahead oil futures curve: the December 1991 futures contract rose only about 4 dollars per barrel while the spot price rose 25 dollars over the same weeks, indicating that market participants expected the shock to be largely temporary. The documented policy response, in the paper's account, was to hold the interest-rate path close to what it would have been absent the shock, a deliberate, judgment-based deviation from the rule's literal short-run prescription, justified by information (the futures curve) that does not enter equation (1) at all. On the fiscal side, the paper notes that the Gramm-Rudman-Hollings law's fixed deficit targets would have forced offsetting spending cuts as the automatic stabilizers pushed the deficit up, and that a legislative revision to the law was required before those stabilizers could be allowed to operate, an explicit institutional illustration of discretion being required even where a "systematic" fiscal response was, in principle, already the intended policy.

5.4 Case study two: German unification and the 1990 bond market

Ten-year U.S. Treasury yields rose 75 basis points in early 1990, a move that, under a price-responsive policy rule, could be read as evidence of rising inflation expectations calling for tighter policy. The paper argues instead that the rise reflected a real, cross-border shock: German long-term yields rose by even more over the same period, and the West German government fiscal

balance moved from roughly a 0.2 percent-of-GDP surplus in 1989 to a deficit of, in the source text consulted, at least 3 percent of GDP in 1990 (the exact digit following "3" is not legible in the copy of the paper used for this review and is not reproduced as a precise figure here, consistent with not fabricating a value the source does not clearly state). Model simulations, again external to this paper, suggested that an interest-rate increase of about one percentage point was consistent with the plausible magnitude of increased German capital demand associated with unification, which the paper takes as evidence that no additional U.S. inflation-driven tightening was required, since the U.S. long-rate rise could be explained by the German real shock transmitted through integrated capital markets rather than by a U.S.-specific rise in expected inflation.

SUMMARY OF THE TWO CASE STUDIES

In both episodes, the paper's policymaker holds the systematic rule's spirit (respond to genuine, persistent inflation and output news) while overriding its literal short-run prescription on the basis of auxiliary market-based evidence, the oil futures curve in one case, cross-country bond-yield comovement in the other, that lies entirely outside the rule's own two arguments.

6. Robustness and Later Evidence

The paper itself contains no formal robustness section: no alternative sample, no alternative functional form estimated and compared, no standard errors, and no out-of-sample test of equation (1). The rule's coefficients are chosen for exposition, and the six-year window used to judge the rule's fit is also the window over which the rule was constructed with the Federal Reserve's actual conduct already in view, so the fit reported in Section 5.2 is, at minimum, partly in-sample by construction rather than an independent check. **CRITIQUE** This is the paper's most consequential methodological gap, and it is one the later literature has addressed directly, even though those later results are not part of the paper under review.

Table 2. Concerns raised by the paper's method and how later literature has addressed them

Concern	Later test	Result	Does it resolve the concern?
Coefficients are calibrated, not estimated; no determinacy criterion is offered for why the inflation coefficient should exceed one	Clarida, Gali and Gertler (2000) estimate forward-looking reaction functions for the pre-Volcker and Volcker-Greenspan periods and connect the inflation coefficient to rational-expectations determinacy	The estimated inflation coefficient is below one (indeterminate, destabilizing region) before 1979 and above one after, consistent with the Taylor principle implicit in equation (1)	Partially. It supplies the missing theoretical and econometric grounding, but for a different, longer sample than Taylor's 1987–1992 window, so it does not directly validate Figure 1 itself
Figures 1–3 are built on revised, ex post National Income and Product Accounts data that the Federal Reserve did not have in real time	Orphanides (2001) reconstructs Taylor-type rule prescriptions using real-time, as-published data vintages instead of revised data	Real-time and revised-data prescriptions diverge substantially, particularly around output-gap estimates, changing conclusions about how "accommodative" or "tight" historical policy appears to have been	No. It shows the specific empirical method used in Section 5.2 of this paper is vulnerable to look-ahead bias; the paper's own visual fit has not, to this reviewer's knowledge, been re-run with real-time vintages for exactly its stated 1987–1992 sample

Source: Clarida, Gali and Gertler (2000); Orphanides (2001), as summarized above; both are later, independent papers, not part of Taylor (1993).

7. Critical Evaluation

7.1 What the paper actually establishes

ESTABLISHED

A simple linear interest-rate rule with coefficients of roughly 0.5 on the inflation gap and 0.5 on the output gap, around a 2 percent real-rate anchor, visually tracks the federal funds rate over 1987–1992 outside the October 1987 stock-market-crash episode.

SUPPORTED BUT CONDITIONAL

Interest-rate rules that respond to price level and output dominate money-supply or exchange-rate rules for joint output-price stability. This rests entirely on the nine-model Bryant-Hooper-Mann comparison and Taylor's own multicountry model, neither reproduced in this paper, so the finding is conditional on model specifications and shock distributions this paper's reader cannot inspect.

SUGGESTIVE

That rule-like, systematic policy conduct actually improved U.S. macroeconomic performance during 1987–1992 relative to a counterfactual discretionary path. The paper cites the general time-inconsistency case for rules (Kydland and Prescott, 1977; Barro and Gordon, 1983) but performs no welfare comparison, counterfactual simulation, or performance metric of its own for the specific 1987–1992 U.S. episode.

NOT ESTABLISHED

That the Federal Reserve was, in any intentional or causal sense, following equation (1) or anything resembling it; that the specific coefficients (0.5, 0.5, a 2 percent inflation target) are optimal rather than illustrative; and that the fit demonstrated in Figure 1 would survive real-time, rather than revised, data.

7.2 Strengths

Three strengths are worth separating from the paper's own stated contributions. First, independently of the author's framing, the specific functional form proposed proved to be an unusually tractable and portable summary of monetary policy conduct: its subsequent adoption by central banks, financial-market economists, and textbooks over the following three decades is itself evidence, **INFERENCE** not asserted by the 1993 paper, that the equation captured something durable about how modern central banks in fact behave, independent of whether Taylor's own justification for the specific coefficients was rigorous. Second, the design/transition/operation trichotomy is a genuinely useful and under-theorized conceptual contribution: the paper is explicit, and correct, that economists are generally better at identifying which policy regime is best than at analyzing the transition toward it, an observation the paper extends by analogy to trade liberalization and market-economy transitions, areas where the same gap exists. Third, grounding the discussion in two real historical episodes, rather than leaving the discretion-preserving argument purely abstract, forces the paper to confront exactly the kind of judgment calls its central thesis is about, which a purely theoretical treatment would not.

7.3 Weaknesses

Independently of the limitations the paper itself acknowledges (chiefly, that mechanical rule-following is impractical), four further weaknesses can be identified. First, because the coefficients in equation (1) are calibrated rather than estimated, the close visual fit in Figure 1 cannot be treated as statistical evidence of anything; at most it demonstrates plausibility. Second, using trailing four-quarter inflation as a proxy for expected inflation is asserted, not tested, and sits awkwardly next to the rational-expectations framework invoked to justify the underlying multicountry simulation evidence. Third, as already noted, the multicountry simulation evidence for the paper's strongest claim (price/output rules dominate money- or exchange-rate rules) is entirely outsourced to an external, unreproduced monograph. Fourth, both case studies are narrative rationalizations constructed with the benefit of hindsight, each resting on a single auxiliary data source (the oil futures curve; German-U.S. bond-yield comovement) selected after

the fact to support a particular reading; a real-time policymaker facing the same events, with the data vintages actually available at the time (the concern later formalized by Orphanides, 2001), might not have been able to draw the same conclusions with the same confidence.

The single most important weakness is a form of circularity: the paper never cleanly separates the rule as a positive description of policy conduct that was, in part, already judged to be good (the coefficients and the sample window are chosen with 1987–1992 Federal Reserve behavior already in view) from the rule as an independently derived normative prescription. Because both objects share the same equation and the same paper, the strong visual fit reported in Section 5.2 risks being, at least in part, evidence for a claim it was partly built to fit.

8. Contradictions and Edge Cases

Several regions of the parameter space and institutional setting expose limits of the mechanism that the paper does not discuss.

8.1 The zero lower bound

Equation (1), $r = 1.5p + 0.5y + 1$, contains no floor. **EXTENSION** Setting $r = 0$ and $y = 0$ requires $p = -2/3$ percent, that is, mild deflation of roughly two-thirds of a percentage point, before the rule alone, from the inflation term, would prescribe a negative nominal rate; a negative output gap would bring the rule to zero even sooner. This region was not empirically relevant in the paper's own 1987–1992 sample, in which inflation stayed positive throughout, but it became centrally important after the 2008 financial crisis and again in 2020, when "shadow" Taylor-rule calculations for the United States, the euro area, and Japan implied deeply negative rates that no conventional interest-rate instrument could deliver. This is a genuine edge case under which the paper's linear mechanism breaks down without modification, and the paper offers no forward-guidance, balance-sheet, or floor-based alternative.

8.2 Fixed exchange-rate regimes

The paper's own multicountry comparison acknowledges that a single-country version of the rule cannot be applied unmodified once exchange rates are fixed, since only one interest rate, a common "world" rate, can then be chosen, and individual countries lose the ability to set policy according to their own domestic price level and output gap. This is not a contradiction the paper overlooks; it is explicitly noted, and it usefully bounds the domain over which the illustrative rule, as stated, is meaningful.

8.3 Identification of the 1987–1992 fit

CRITIQUE Could another simple rule generate the same qualitative fit to 1987–1992 Federal Reserve behavior? Plausibly yes. U.S. monetary policy over this specific window was dominated by a single ongoing disinflation (the tail of the Volcker-Greenspan program) and two clearly dated recessions, so almost any rule that is mildly procyclical in the interest rate and anti-inflationary in direction

would broadly track the same path over six years. The paper does not compare equation (1) against McCallum's (1988) rival base-growth rule, or against a pure inflation rule, on the same sample, so the specific two-gap functional form is not uniquely identified by the evidence the paper presents; the fit shown is consistent with equation (1) without being distinctive to it.

8.4 High-inflation extremes

At the opposite extreme, if inflation were to rise to, say, 10 percent against the paper's 2 percent target, equation (1) prescribes $r=1.5(10)+1=16$ percent nominal (holding $y=0$), a level not historically unreasonable given that the effective federal funds rate approached 20 percent during the Volcker disinflation of 1981. Unlike the zero-lower-bound region, the rule's linear form does not obviously misbehave at high inflation, which suggests the mechanism's practical fragility is asymmetric: more exposed to the low-inflation, low-output-gap region than to the high-inflation region.

9. Further Literature

Kydland and Prescott (1977). Established the formal time-inconsistency case for rules over discretion. This paper is best read as a direct attempt to operationalize that result into a specific, usable interest-rate rule, without re-deriving or re-testing the underlying game-theoretic argument.

Barro and Gordon (1983). Supplies the reputational, repeated-game formalization of the same rules-versus-discretion distinction that Taylor cites as one of three equivalent framings (alongside Kydland-Prescott and Blanchard-Fischer) motivating the paper's premise that rule-like behavior is desirable.

McCallum (1988). Proposes a rival simple rule based on monetary-base growth. Cited by Taylor as part of the same wave of policy-rule research but never numerically compared against equation (1) within this paper, leaving an open empirical question this review returns to in Section 10.

Taylor (1979, 1993 book). Supplies essentially all of the structural, estimated multicountry model evidence this paper summarizes qualitatively. The 1993 conference paper under review is best understood as a non-technical companion piece to the 1993 book, not as a self-contained empirical contribution.

Bryant, Hooper and Mann (1993). The independent nine-model comparison that supplies the cross-model consensus claim discussed in Section 5.1. This is complementary rather than contradictory evidence, since it was built by institutions other than Taylor and arrives, largely, at the same qualitative ranking of policy rules.

Clarida, Gali and Gertler (2000). Gave equation (1) its later theoretical foundation by connecting the size of the inflation-response coefficient to rational-expectations determinacy in a New Keynesian model, and by estimating, rather than calibrating, the coefficient for both a pre-Volcker and a Volcker-Greenspan sample. This is the natural theoretical and econometric extension of the present paper, addressing precisely the gap identified in Section 7.3.

Orphanides (2001, 2003). Shows that Taylor-type rule prescriptions, and conclusions about historical Fed behavior drawn from them, change substantially when real-time rather than revised data are used. This is a direct methodological challenge to the empirical method used in Section 5.2 of the present paper.

Woodford (2001). Provides the micro-founded, Euler-equation derivation of why an interest-rate rule needs an inflation-response coefficient above one for a determinate, stable rational-expectations equilibrium, formalizing what the present paper only discovers, unremarked, in its own arithmetic.

10. Research Gaps

Paper-Level Gaps

- No econometric estimation of the coefficients in equation (1) is performed within this paper; the rule is calibrated, not fitted.
- The two case studies are ex post narrative rationalizations without a pre-specified decision rule for when a policymaker should override the equation.
- The multicountry simulation evidence underlying the paper's strongest claim is entirely external and unreproduced.
- No treatment of the zero or effective lower bound on nominal interest rates.

Literature-Level Gaps

- How should a real-time policymaker distinguish a temporary, futures-market-confirmed shock from a genuinely persistent one, without the benefit of hindsight available to a 1993 paper written after the events it discusses?
- How should a simple rule of this form be adapted for a single small open economy operating a managed, rather than purely fixed or purely flexible, exchange-rate regime, an intermediate case the paper's binary fixed/flexible comparison does not address?

Methodological Gaps

- Better, transparently documented real-time trend-output and output-gap estimates; the paper's 2.2 percent trend growth assumption is stated without a construction method.
- Formal determinacy analysis of the rule within a rational-expectations model, only supplied years later by Clarida, Gali and Gertler (2000) and the New Keynesian literature.
- An explicit welfare-loss comparison of the illustrative rule against both pure discretion and McCallum's (1988) rival rule, evaluated on the same 1987–1992 sample.

11. Research Opportunities

1. A real-time replication of Figure 1

RESEARCH QUESTION: Does the rule's apparent fit to 1987–1992 Federal Reserve conduct survive once real-time, rather than revised, output-gap and inflation data are used?

MOTIVATION: The entire empirical case in Section 5.2 rests on data the Federal Reserve could not have observed at the time.

HYPOTHESIS: The real-time fit is visibly worse than the revised-data fit, particularly around the 1990–91 recession, when output-gap estimates were revised substantially.

ECONOMIC MECHANISM: Statistical agencies revise GDP and price data for years after first release; a policymaker in 1990 saw different numbers than those Figure 1 plots.

LITERATURE CONNECTION: Orphanides (2001).

DATA: Philadelphia Federal Reserve Real-Time Data Research Center vintage series for real GDP, the GDP deflator, and potential output.

VARIABLES: Vintage-specific real GDP, GDP-deflator inflation, potential-output estimates.

METHOD: Reconstruct the rule-implied rate quarter by quarter using only the data vintage available at each FOMC meeting date, and compare to Taylor's revised-data plot.

IDENTIFICATION STRATEGY: Within-quarter data-vintage variation; not a regression-based identification strategy.

EXPECTED CONTRIBUTION: A direct, quantified measure of how much of the paper's claimed good fit is a real-time illusion.

MAIN RISK: Real-time vintage data for the late 1980s may be incomplete or unavailable at sufficient quarterly resolution for a fully clean replication.

2. An updated, zero-lower-bound-aware cross-rule model comparison

RESEARCH QUESTION: Does the paper's qualitative ranking of interest-rate rules above money-growth rules survive in modern models and a sample that includes the 2008–2015 and 2020–2021 near-zero-rate episodes?

MOTIVATION: The nine-model comparison behind the paper's strongest claim used 1990s-vintage models and a pre-crisis sample.

HYPOTHESIS: Interest-rate rules continue to dominate simple money-growth rules for output and price stability in modern dynamic stochastic general equilibrium models, but the margin narrows once the effective lower bound binds.

ECONOMIC MECHANISM: Money-growth rules face no analogous lower bound, so their relative disadvantage may shrink exactly when interest-rate rules are most constrained.

LITERATURE CONNECTION: Bryant, Hooper and Mann (1993); Taylor (1993).

DATA: Model-simulated data; historical shock distributions re-estimated over 1990–2025.

VARIABLES: Output gap, inflation, short-term policy rate, money growth.

METHOD: Stochastic simulation under alternative rules, following the original Bryant-Hooper-Mann design.

IDENTIFICATION STRATEGY: Counterfactual model simulation holding structure fixed across policy-rule regimes.

EXPECTED CONTRIBUTION: An updated, lower-bound-aware version of the paper's central cross-rule ranking.

MAIN RISK: Most modern central banks have abandoned monetary targeting, making a genuinely comparable money-growth rule somewhat anachronistic to construct.

3. Formal identification of the two 1990 case studies

RESEARCH QUESTION: Would a modern identified-shock methodology validate Taylor's narrative judgment that the 1990 oil shock was temporary and the 1990 bond-yield rise was a real, not inflationary, disturbance?

MOTIVATION: Both case studies are argued narratively, with tools that can now be formalized.

HYPOTHESIS: The oil futures curve's slope in August-October 1990 significantly forecasts the subsequent path of U.S. core inflation, and cross-country long-bond comovement with Germany explains most of the 1990 U.S. long-rate rise.

ECONOMIC MECHANISM: Futures prices aggregate market expectations of shock persistence; cross-country yield comovement identifies a common real shock as distinct from a country-specific inflation shock.

LITERATURE CONNECTION: Kilian (2009) on oil-shock decomposition; the term-structure decomposition literature.

DATA: 1990 NYMEX and IPE oil futures curves; U.S. and German ten-year bond yields; U.S. core inflation.

VARIABLES: Futures-implied expected spot price, term spread, realized subsequent inflation.

METHOD: Event-study local projections around the two episodes, or a small identified vector autoregression with sign restrictions.

IDENTIFICATION STRATEGY: Futures-curve slope and cross-country yield comovement as external instruments for the "temporary" and "real" shock interpretations, respectively.

EXPECTED CONTRIBUTION: Converts Taylor's narrative case studies into testable, replicable claims.

MAIN RISK: Two episodes provide very few effective degrees of freedom; results would be suggestive at best.

4. Formal estimation of the original 1987–1992 rule

RESEARCH QUESTION: If Taylor's rule is estimated, rather than calibrated, on exactly his stated sample, are the estimated coefficients statistically distinguishable from his assumed 0.5 and 0.5, and does the estimated inflation coefficient exceed the Taylor-principle threshold of one?

MOTIVATION: Bridges the calibration-only nature of the 1993 paper with the estimation methodology Clarida, Gali and Gertler later applied to a longer sample.

HYPOTHESIS: The estimated inflation coefficient for exactly the 1987–1992 window is close to, but not statistically indistinguishable from, Taylor's assumed total response of 1.5.

ECONOMIC MECHANISM: Generalized-method-of-moments estimation of a backward- or forward-looking interest-rate reaction function.

LITERATURE CONNECTION: Clarida, Gali and Gertler (2000).

DATA: The same 1987–1992 window, in both revised and real-time vintages.

VARIABLES: Federal funds rate, expected or lagged inflation, output gap, lagged funds rate for interest-rate smoothing.

METHOD: Generalized method of moments with lagged macroeconomic instruments, following Clarida, Gali and Gertler's methodology.

IDENTIFICATION STRATEGY: Orthogonality of instruments to the reaction-function residual.

EXPECTED CONTRIBUTION: A formal, replicable estimate of the exact rule the paper claims to have found only visually.

MAIN RISK: Roughly twenty-four quarterly observations is a very short sample for generalized-method-of-moments estimation; standard errors are likely to be large.

5. Testing whether publishing Taylor-rule benchmarks changed central-bank behavior

RESEARCH QUESTION: Do central banks that publish Taylor-rule-style benchmarks, as the Federal Reserve has done in its Monetary Policy Report since 2017, show different policy-rate volatility or forecast-error properties than central banks that do not?

MOTIVATION: Taylor's own proposed operational mechanism, adding rule comparisons to the FOMC briefing materials, was in fact adopted decades after this paper was written.

HYPOTHESIS: Publication of Taylor-rule benchmarks is associated with smaller policy-rate reversals and smaller inflation-forecast errors relative to a matched sample of central banks without such publication.

ECONOMIC MECHANISM: Public benchmarking increases communication clarity and anchors expectations, consistent with the credibility mechanism the paper borrows from the time-inconsistency literature.

LITERATURE CONNECTION: Kydland and Prescott (1977); the central-bank transparency and communication literature.

DATA: Federal Reserve Monetary Policy Report Taylor-rule tables (2017 onward); matched central banks (European Central Bank, Bank of England, Bank of Canada) without equivalent published tables.

VARIABLES: Policy-rate changes, inflation and output forecast errors, the gap between rule-implied and actual rates.

METHOD: Difference-in-differences around the Federal Reserve's 2017 adoption date, or a panel regression across central banks with a communication-practice indicator.

IDENTIFICATION STRATEGY: Staggered adoption of Taylor-rule publication across otherwise similar inflation-targeting central banks.

EXPECTED CONTRIBUTION: Direct evidence on whether the paper's operational proposal, once actually implemented, delivers the credibility benefit its theoretical framing predicts.

MAIN RISK: Very few central banks publish comparable tables, so the cross-sectional sample is small and confounded by many other differences between central banks.

Ranking of the Strongest Ideas

Idea 1 ranks first: it is the most feasible, since the Philadelphia Federal Reserve's real-time dataset is a mature and well-documented public resource; it carries the highest methodological rigor relative to the effort required; and it directly and cleanly tests the paper's single most load-bearing empirical claim. Idea 4 ranks second: the short-sample problem, honestly flagged as its main risk, is real, but it is the most direct econometric bridge between Taylor's calibration-era rule and Clarida, Gali and Gertler's estimation-era methodology, and no study appears to have re-estimated exactly Taylor's stated sample and functional form. Idea 3 ranks third: it extends the paper's own most distinctive contribution, the two grounded case studies, but with only two historical episodes its statistical power is inherently limited, placing it behind the two data-rich alternatives. Ideas 2 and 5 rank lower, the former because reproducing a nine-model comparison exceeds the scope of a single research project, the latter because the cross-section of central banks publishing comparable rule benchmarks is small and confounded.

12. Learning Framework

Prerequisite Concepts

A working sense of the output gap and potential output; rational expectations and the Lucas critique; the time-inconsistency and credibility problem in a repeated policy game; and enough open-economy intuition (interest parity) to see why a fixed exchange rate removes a country's ability to set its own interest rate independently.

Core Concepts Learned

The design, transition, and operation trichotomy for policy rules; the specific two-gap linear interest-rate reaction function later known as the Taylor rule; the distinction between a policy rule as a mechanical algorithm and as a "systematic," credible posture; and how rational-expectations model simulation, rather than reduced-form regression, is used to evaluate policy rules once the Lucas critique is taken seriously.

Important Equations

Equation (1), $r = p + 0.5y + 0.5(p - 2) + 2$, equivalently $r = 1.5p + 0.5y + 1$, and the output-gap definition $y = 100(Y - Y^*)/Y^*$.

Methods Learned

Calibrating, as distinct from estimating, a policy reaction function; using a futures curve to separate a temporary from a permanent shock; and using cross-country long-bond comovement to distinguish a common real shock from a country-specific inflation shock.

Questions to Test Understanding

Comprehension. In one sentence, why does Taylor argue a central bank should not follow equation (1) mechanically even while believing rule-like behavior improves policy performance? What distinguishes, analytically, a "transition" between policy rules from ordinary discretionary policy?

Analysis. Why does the identification of the claim that price/output interest-rate rules dominate money- or exchange-rate rules rest on rational-expectations model simulation rather than a historical regression, and what does the Lucas critique have to do with that choice? Starting from equation (1), derive the simplified form $r = 1.5p + 0.5y + 1$ and explain why the 1.5 coefficient on inflation implies that the real interest rate rises whenever inflation rises.

Research. How would you test, using data unavailable to Taylor in 1993, whether the Federal Reserve was actually following something like equation (1) during 1987–1992, rather than the rule merely resembling Fed conduct after the fact? Design a single empirical test that would discriminate between the paper's claim that the 1990 bond-yield rise reflected German-unification-driven real demand rather than rising U.S. inflation expectations.

Papers to Read Next

Kydland and Prescott (1977); Clarida, Gali and Gertler (2000); Orphanides (2001); Taylor (1993), *Macroeconomic Policy in a World Economy*; McCallum (1988).

13. Conclusion

One-Sentence Thesis

Taylor operationalizes the abstract rules-over-discretion result of the credibility literature into a simple, calibrated interest-rate reaction function, arguing that such a rule should guide rather than mechanically dictate central-bank decisions, and shows, largely through narrative rather than statistical evidence, that recent Federal Reserve conduct and two 1990 policy episodes are broadly consistent with that posture.

Three Most Important Insights

A policy rule can be systematic and credible without being mechanically executed, an idea that dissolves much of the old rules-versus-discretion dichotomy. The specific two-gap functional form of equation (1) proved to be an unusually durable and portable summary of monetary policy conduct, later given rigorous theoretical grounding by the New Keynesian determinacy literature this paper does not itself contain. Using a rule to structure judgment, as in the two case studies, requires exactly the kind of auxiliary, real-time information, futures prices, cross-country yield comovement, that the rule's own two arguments do not carry.

Strongest Evidence

The qualitative cross-model finding that price/output-responsive interest-rate rules dominate money- or exchange-rate rules for joint output-price stability, because it is corroborated across several independently specified rational-expectations models (Bryant, Hooper and Mann, 1993) rather than resting on Taylor's framework alone.

Biggest Weakness

The central visual evidence for equation (1), Figure 1's fit to 1987–1992 Federal Reserve behavior, is calibrated rather than estimated, built on revised rather than real-time data, and never benchmarked against any rival simple rule, so it cannot bear the weight later literature and central-bank practice have placed on it.

Most Important Research Gap

No version of the rule addresses the possibility that the prescribed nominal rate falls below zero, a scenario central to monetary policy for over a decade following the 2008 financial crisis and again in 2020.

Best Research Opportunity

The real-time replication of Figure 1 (Research Idea 1), because the necessary data now exists, the test is decisive for the paper's core empirical claim, and no comparable exercise appears to have been carried out for exactly Taylor's stated sample and rule.

Three Papers to Read Next

Kydland and Prescott (1977), "Rules Rather than Discretion: The Inconsistency of Optimal Plans"; Clarida, Gali and Gertler (2000), "Monetary Policy Rules and Macroeconomic Stability: Evidence and Some Theory"; Orphanides (2001), "Monetary Policy Rules Based on Real-Time Data."

14. References

- Barro, R.J. and Gordon, D.B. (1983). Rules, Discretion and Reputation in a Model of Monetary Policy. *Journal of Monetary Economics*, 12, 101–122.
- Blanchard, O. and Fischer, S. (1989). *Lectures in Macroeconomics*. Cambridge, Massachusetts: MIT Press.
- Bryant, R., Hooper, P. and Mann, C. (1993). *Evaluating Policy Regimes: New Research in Empirical Macroeconomics*. Washington, D.C.: Brookings Institution.
- Clarida, R., Gali, J. and Gertler, M. (2000). Monetary Policy Rules and Macroeconomic Stability: Evidence and Some Theory. *Quarterly Journal of Economics*, 115(1), 147–180.
- Economic Report of the President (1990). Washington, D.C.: United States Government Printing Office.
- Kilian, L. (2009). Not All Oil Price Shocks Are Alike: Disentangling Demand and Supply Shocks in the Crude Oil Market. *American Economic Review*, 99(3), 1053–1069.
- Kydland, F. and Prescott, E.C. (1977). Rules Rather than Discretion: The Inconsistency of Optimal Plans. *Journal of Political Economy*, 85, 473–492.
- Laidler, D. (1993). Price Stabilization in the 1990s: A Summary, in K. Shigehara (ed.), *Price Stabilization in the 1990s*. London: Macmillan Press.
- McCallum, B. (1988). Robustness Properties of a Rule for Monetary Policy. *Carnegie-Rochester Conference Series on Public Policy*, 29, 173–203.
- Orphanides, A. (2001). Monetary Policy Rules Based on Real-Time Data. *American Economic Review*, 91(4), 964–985.
- Orphanides, A. (2003). Historical Monetary Policy Analysis and the Taylor Rule. *Journal of Monetary Economics*, 50(5), 983–1022.

- Taylor, J.B. (1975). Monetary Policy During a Transition to Rational Expectations. *Journal of Political Economy*, 83, 1009–1021.
- Taylor, J.B. (1979). Estimation and Control of a Macroeconomic Model with Rational Expectations. *Econometrica*, 47, 1267–1286.
- Taylor, J.B. (1993). *Macroeconomic Policy in a World Economy: From Econometric Design to Practical Operation*. New York: W.W. Norton.
- Taylor, J.B. (1993). Discretion versus Policy Rules in Practice. *Carnegie-Rochester Conference Series on Public Policy*, 39, 195–214.
- Woodford, M. (2001). The Taylor Rule and Optimal Monetary Policy. *American Economic Review*, 91(2), 232–237.